Project Report

On

College Club Management System

Submitted as partial fulfilment for the award of

BACHELOR OF TECHNOLOGY DEGREE

SESSION 2025 – 2026

In

Computer Science Engineering (AI)

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(Formerly UPTU)

May, 2026

DECLARATION

We certify that the submission is our own work that, so far as we know or believe, it comprises no work that has previously been published or written by some other person or work which to any substantial degree has been considered in the award of any other degree or diploma of the university, or any other institution of higher learning, except that due acknowledgment has been made in the work itself.

We also state that we have conducted the project under the guidance and supervision of Mr. Mukesh Lomror in the last year of the Bachelor of Technology degree in Computer Science and Engineering (Artificial Intelligence) on the subject matter of the project, i.e., the management of college clubs in a web-based model of greater interaction with students and organization of events.

This is not something that has been turned to any other place to fulfill the need of a course or degree/diploma in this or any other institution.

We also state that we mention all the information and materials we have applied in this project and this has been credited.

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CERTIFICATE

This is to certify that the project report entitled "College Club Management System: A Web-Based Platform for Enhanced Student Engagement and Event Coordination" submitted by Anuj Gupta, Vinit Vishwakarma, Vineet Singh, and Vipin Yadav in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence) at KIET Group of Institutions, Ghaziabad, is a record of the candidate's own work carried out by them under my supervision and guidance.

The work embodied in this report is original and has not been submitted, either in part or in full, for the award of any other degree or diploma in this or any other institution.

The project demonstrates the development and implementation of a web-based system using modern technologies such as the MERN stack, which includes features like club management, membership handling, event coordination, and communication in an efficient and structured manner. The system has been tested and found to be working satisfactorily.

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B.Tech. Final Year Project undertaken during our final year. We express our sincere gratitude to our project guide **Mr. Mukesh Lomror**, Assistant Professor, Department of Computer Science & Engineering (Artificial Intelligence), KIET Group of Institutions, Ghaziabad, for his invaluable guidance, continuous support, and encouragement throughout the development of this project. His insights, suggestions, and dedication have played a crucial role in the successful completion of this work.

We are also thankful to all the faculty members of the Department of Computer Science & Engineering (AI), KIET Group of Institutions, Ghaziabad, for their constant support and cooperation. Their guidance and motivation helped us overcome various challenges during the course of this project.

We would also like to acknowledge our friends and classmates for their valuable suggestions, encouragement, and support, which contributed significantly to the completion of this project.

Finally, we express our heartfelt gratitude to our parents and family members for their continuous support, motivation, and encouragement, which have been a constant source of strength throughout this work.

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ABSTRACT

Among a few of the challenges experienced in the management of college clubs are witnessing delays in the approval process, poor communication, and poor record-keeping. The poor coordination between students, faculty members, and administrators is caused by traditional means such as using manual processes and fragmented digital tools.

This project aims to propose solutions to these issues with the help of a college club management system, a web-based centralized website to make operation of various operations related to clubs quick and automated. The system combines the essential features, including role-based access control, membership management, scheduling events, document management, and intra-communication, into one platform.

Modern web technologies (MERN stack) are used to develop the application, guaranteeing a responsive user interface and a stable system on the back-end. It employs MongoDB to store data efficiently and Node.js and Express.js to carry out operations on the server. An interactive front-end is built using React.js that improves user experience. The system has a variety of roles of users, such as administrators, members of the faculty, club heads, and students with distinct functionalities and access control.

Also, the system has a built-in AI-powered chatbot, which will act as the assistant system, responding to queries immediately and suggesting appropriate clubs depending on the preferences of the user. The suggested system will handle the drawbacks of the old systems by strengthening the efficiency of the workflow, increasing its transparency, and coordinating all stakeholders more effectively

The system has been tested on its functionality, usability, and performance. It is shown to be much more efficient, better at communication, and capable of data management than the current interface based on the manual and semi-digital systems. All in all, the system reinvents the traditional club management system into a streamlined and easy-to-scale digital solution, which enhances a greater number of students to participate in and efficiently manage the activities of the college clubs.

Keywords: College Club Management, Role-Based Access, Event Coordination, Membership Management, MERN Stack, AI Chatbot, Web-Based System.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
MERN	MongoDB, Express.js, React.js, Node.js
API	Application Programming Interface
DBMS	Database Management System
UI	User Interface
UX	User Experience
JWT	JSON Web Token
HTTP	Hyper-Text Transfer Protocol
HTTPS	Hyper-Text Transfer Protocol Secure
CRUD	Create, Read, Update, Delete
NoSQL	Not Only SQL (Non-relational Database)
HTML	Hyper-Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
REST	Representational State Transfer
RBAC	Role-Based Access Control
LMS	Learning Management System

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF COLLEGE CLUB MANAGEMENT SYSTEM

Technology is quite significant in the current digital age in terms of enhancing efficiency and management of various kinds of activities. Nearly all spheres, such as education, business, healthcare, and communication, require digital systems to carry out their tasks in the fastest, most accurate, and well-structured way possible. Colleges and universities are also utilizing the use of digital solutions to enhance academic and non-academic operations. Management of college clubs and student affairs is one of the key areas where a digital transformation is required.

College clubs are an essential part of a student's life. They offer chances to students to engage in their interests outside the classroom. In such clubs, students are allowed to take part in most activities, including the technical events, cultural programs, sports competitions, workshops, and social services. These activities do not only render college life more interesting but also enable the students to acquire vital skills. These enhance leadership skills, teamwork skills, communication skills, and problem-solving skills among students that come in handy during their future lives.

Moreover, college clubs provide the students with a chance to communicate with other people, exchange ideas, and get to know new things. They also enable students to develop self-confidence and enhance their character. Being involved in club activities gives the students a chance to have hands-on experience in addition to what they learn in school. Thus, these clubs need proper maintenance to be managed well and allow maximum participation of students.

Nevertheless, in most colleges, running clubs is still carried out in the old-fashioned modes of using paperwork, emails, spreadsheets, and informal channels of communication. Such approaches are not effective and pose various difficulties. Paper-based records and unnecessary management take time and introduce the likelihood of errors. Data can be lost or misplaced, potentially containing important information, and this concerns the reliability of the system.

Delay in process, e.g., membership approvals and event management, is one of the most striking issues with traditional systems. These processes rely on manual communication and are therefore time-consuming and could be inconvenient to the users. The students are not updated in time on whether they are members or not and on upcoming events, and the lowest participation is lessened.

Poor communication is another problem. Without a centralized system of communication, crucial announcements are frequently relayed on various platforms, including emails or Unity messaging software. This renders cases where the users might not be able to keep up due to the information not being received by everyone at the right time. Consequently, communication between students, faculty, and administrators is impaired.

Another huge issue in conventional systems is that of data management. Storing data manually or on more than one platform makes maintaining consistency and accuracy challenging. Membership and event records as well as communication records might not be arranged appropriately. This hinders availability of information when required and heightens chances of loss of data.

The issues are worse in high-activity events like college fests, workshops, and orientation programs. It is highly challenging to control two or more clubs and events simultaneously without an effective system. Progress or activity tracking is not well-organized and makes people confused, resulting in inefficiency.

Most of the digital tools are limited in functionality, even though there are a few available. They are not meant to deal with everything about the club management under a single roof. Consequently, users have to continue relying on several systems, which make them more complex and less efficient.

In order to deal with these obstacles, a centralized and effective system would be required that can be utilized to handle all activities that are related to the club on a single platform. This need is met by the proposed College Club Management System. It offers one platform where all processes and activities like membership, event planning, communication, and data handling would be handled easily.

This system enhances greater transparency, manual work minimization, and real-time updates. It maintains all users connected and updated. On the whole, it helps to streamline the organization of college clubs, make this task more structured, and get more students involved in extracurricular activities. This system is especially useful during high activity periods such as college fests, where manual systems fail to handle large volumes efficiently.

1.2 PROJECT DESCRIPTION

The College Club Management System is a web-based application developed to automate and simplify the management of college clubs. The main objective of this system is to provide a centralized platform where all club-related activities can be managed efficiently.

The system allows students to register, explore different clubs, and join them based on their interests. Students can also participate in various events organized by the clubs. At the same time, administrators and club heads can manage memberships, organize events, and communicate with users in a structured manner. This reduces manual work and makes the entire process more efficient.

The application is developed using modern web technologies, specifically the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. MongoDB is used for storing data in a flexible and structured format. Node.js and Express.js are used for backend development, where all server-side operations and API handling are performed. React.js is used for frontend development to create an interactive and user-friendly interface.

The system follows a client-server architecture, where the frontend and backend communicate through APIs. The frontend sends requests to the backend, and the backend processes these requests and returns the required data. This ensures fast performance and smooth operation of the system.

One of the key features of this system is role-based access control. Different users have different roles, such as Admin, Faculty, Club Head, and Member. Each role has specific permissions and responsibilities. For example, the admin manages the entire system, faculty members monitor activities, club heads manage members and events, and students can join clubs and participate in events. This ensures security and proper control over system operations.

The system includes several important features such as club registration, membership management, event scheduling, notifications, and communication tools. Students can easily browse available clubs and join them. Club heads can create and manage events, while users receive updates through notifications.

All data is stored in a structured format in the database, making it easy to access and update information. The system also supports real-time updates, ensuring that users always have access to the latest information.

Overall, the system reduces manual effort, improves communication, enhances transparency, and provides a scalable and reliable solution for managing college club activities.

1.3 OBJECTIVES

The primary aims of the suggested College Club Management System are the following:

• To establish a central focused space that will allow all club activities to be centralized and, therefore, easier to manage and access.

• To decrease manual labor through automation of activities like approval of memberships, event management, and communications.

• To enhance the overall efficiency of operations through enhancing the simplicity of workflow and shortening delays due to approvals and updates.

• To strengthen interaction among the students, faculty members, and administrators with a systematic and dependable framework.

• The First Place: To guarantee the accuracy of data and its openness, store all data in a well-organized database, and minimize the risks of losing data.

• To offer real-time updates to ensure that users can have access to the current information regarding clubs, events, and membership.

• To enhance coordination between various users by giving clear roles and responsibilities in the system.

• To make the interface user-friendly and simple to operate, even with simple technical skills.\

• To make more students participate and engage in the activities of the club by simplifying how students can explore, join, and take part in club activities.

1.4 ORGANIZATION OF THE REPORT

These report sections have been divided into five chapters so as to have a clear and orderly description of the project. The system is divided into different aspects that are covered in each chapter, and it is easier to follow how the whole workflow and implementation are.

Chapter 1 deals with the general idea of the project, such as the background, problem statement, and objectives. It describes the reasons why a system like it is required and what the project will accomplish.

Chapter 2 includes the review of the existing systems, as well as the related work in detail. It examines the solutions that are available in the present market and reveals their weaknesses, which contributes towards recognition of the requirement of the proposed system.

Chapter 3 outlines the suggested approach that will be adopted by the project. It comprises system architecture, workflow, and user roles as well as technical details. This chapter describes the design of the system and its functionality.

Chapter 4 talks about the outcomes of the system and gives an analysis of its performance. It demonstrates the ability of the system to be more efficient in comparison with the more traditional approaches and outlines major results.

The report ends with the conclusion of chapter 5, which summarizes the whole work on the project. It also gives the future outlook, how they can be improved, and what more features they can have to be developed further.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The literature review is an important constituent of this project since it gives a holistic perspective of the systems, technologies, and research available in the field regarding the sphere of college club management. It assists in the analysis of the similar problems in the past and what sort of solutions are being offered now in this field [2], [9]. It is possible to understand the strengths, limitations, and gaps of these systems by studying them, and this is what helps to develop a better and more effective system [3], [7].

College clubs are important in the overall development of students in schools [1]. These clubs offer a place in which students can mainly explore their interests, improve their practical use of knowledge, and engage in several extracurricular activities. Other types of activities like technical, cultural, sports, and social events are known to equip students with the necessary skills like leadership, working in a team, communication, and problem-solving [5]. Thus, a good management of such clubs would be critical in ensuring a qualitative and interactive campus life [6].

Although they are of great importance, the management of college clubs is done in a conventional manner in most institutions [2]. These techniques will consist of paperwork, emails, spreadsheets, and informal interaction through messaging services. Although these methods might be effective in small-scale operations, they turn out to be ineffective and hard to control as the number of clubs, members, and activities grows [8]. Not only are these traditional systems time-consuming, but they are also likely to be affected by human errors, as they influence the overall system performance [10].

Lack of proper organization is one of the biggest problems of manual systems [7]. The details about club memberships and events, as well as communication, are usually dispersed across several sources. This complicates the ensuring of accuracy and consistency in data. Critical facts are widely lost, misplaced, or overlooked, and confusion and inefficiency prevail [8].

Delays in processes like membership approvals and event management are another challenge [4]. Given that these tasks are done manually, a set of steps is needed, and in many cases, processing time is dependent on one-on-one communication, which also increases the time of processing. This leads to lack of user experience and involvement by the students [3].

As technological progress makes headway, there are those institutions that are beginning to embrace the digital tools in the management of club activities [2], [9]. The majority of these tools are, however, basic systems that are underdeveloped and do not go beyond a particular functionality, like event registration or announcement [4]. They fail to offer a holistic solution that incorporates club management in one platform [7].

These systems are not well integrated, posing some challenges [8]. This makes users jump between the use of various platforms to execute various tasks, thus complicating things and diminishing efficiency. In addition, it is challenging to ensure consistency of data in various systems, which causes mistakes and redundancy of data [10].

This literature review aims to examine these current systems in detail and the limitations that they have [3], [7]. The work pays attention to such crucial items as the system of communication, role-based access control, coordination of events, and data management practices. These factors are critical to the proper operational processes of any organizational system [6].

Good communication is a factor that makes the sharing of information between users easy [8]. Role-based access control helps in maintaining security and proper access to system functionalities. Event coordination helps in managing activities efficiently, and proper data management helps in being accurate and reliable [10].

With this type of analysis, we can conclude that the current systems do not provide the capability to support the complex needs of management within the college clubs [2], [9]. Thus, it requires a centralized, integrated, and automated system that is capable of combating these challenges. The purpose of the proposed system is to offer one common platform, which will better organize, increase transparency, and provide control with efficient management of the club activities [7].

2.2 EXISTING SYSTEMS AND ANALYSIS

Existing systems analysis is a significant stage in the understanding of how the existing solutions work and fail [3], [7]. In college club management, there are different methods being employed both in the traditional and online settings [2]. Yet, the majority of these systems have become obsolete or have been developed at the most partial level, restricting their capacity to assist with the increased complexity of activities in clubs [9].

Existing systems may be typically divided into manual techniques, simple digital instruments, and partly automated systems [8]. Both of these solutions offer some functionalities, though neither of them is a full-fledged, built-in solution to managing all aspects of club operations [7]. Consequently, institutions tend to struggle with efficiency, coordination, and data consistency [10].

Existing systems have shown to be very low in the area of integration [8]. Various activities that include registration, communication, event management, and recordkeeping are processed through diverse tools or platforms. This makes the workflow very fragmented, and users may have to navigate across a host of systems to get things done. This decreases efficiency, as it not only adds complexity but also results in work duplication and variations in data [10].

Lack of adequate automation is another problem [4], [7]. Membership approvals, event scheduling, and communication processes, among many others, are yet to be automated or have a high level of human intervention in the process. This causes delays, less efficiency, and increases the possibility of errors [3]. Automation is needed so that more tasks and users can be processed quickly and without errors when the number of activities and users is high [6].

Ineffective communication systems are also highly prevalent in stated systems [8]. In a lot of situations, people conduct communications by emails and messaging platforms that are not connected to the main system. It results in missed updates and slow responses and lack of coordination between students, faculty members, and administrators [2]. It must have a centralized communication system so that users can get the relevant information in time [7].

Moreover, most systems lack the use of an effective role-based access control, which is a critical aspect in ensuring security and efficiency [3]. The lack of access control and defined roles can result in confusion and abuse because, without these, users might be given access to other system features that are not applicable to them. A properly designed role-based system makes sure every user is only able to do what he/she is supposed to do according to his/her role [6].

The other constraint is on data management [10]. In most of the current systems, data remains in unstructured form and/or is spread out across various platforms [8]. This complicates the aspect of consistency and accuracy. It also exposes our systems to risks of losing data and also complicates access to information when required. Data management is crucial in assuring reliability and aiding the decision-making process [10].

More so, scalability is a significant issue when using traditional and partly digital systems [7]. These systems cannot effectively cope with the increasing workload as more and more clubs, members, and events become part of them. This leads to performance problems and lower system performance [4]. The needs of the modern institution must be designed in a manner that they may be easily scaled as per the needs of the institution [6].

The analysis of these existing systems clearly shows that there is a need for a more comprehensive and integrated solution [2], [9]. The efficiency and user experience could be greatly enhanced by using a centralized system that integrates all the functions, like membership management, event classification, communication, and data management, into one platform [7].

The subsections below argue in detail about certain aspects of available systems, their strengths, and their limitations [3]. This analysis assists in knowing the weaknesses that prevail in the existing solutions and the reasons as to why the proposed College Club Management System is necessary [7].

2.2.1 ONLINE NETWORKS FOR STUDENT ACTIVITY MANAGEMENT

Many educational institutions still rely on outdated and basic digital tools for managing student activities [2], [9]. These systems are generally limited to simple functionalities such as registration forms, spreadsheets, or basic data storage. While they provide a starting point for managing information, they lack advanced features required for efficient club management [7].

One of the major issues with these systems is the absence of workflow management [3]. For example, when a student applies to join a club, there is no proper system to track the approval process. This creates confusion for both students and administrators. Students are often unaware of the status of their requests, while administrators struggle to manage multiple applications at the same time [8].

Another limitation is the lack of centralized access to information [8]. Data related to clubs, events, and memberships is often stored in different locations, making it difficult to retrieve and update information. This leads to delays and inefficiency in handling operations [10].

Additionally, these systems are not designed to handle the increasing number of clubs and participants in modern educational institutions [7]. As the scale grows, these tools become difficult to manage and fail to provide the required level of organization [4].

Overall, online networks used for student activity management provide only basic support and are not suitable for handling complex club operations [2]. This highlights the need for a more structured and integrated system [7].

2.2.2 ROLE-BASED ACCESS AND ADMINISTRATIVE WORKFLOWS

Role-based access control (RBAC) is a key requirement for any efficient management system, as it ensures that users have access only to the functionalities relevant to their roles [3], [6]. This improves system security, reduces confusion, and ensures proper management of tasks.

However, many existing systems do not implement RBAC effectively [7]. Instead, they provide uniform access to all users, which leads to inefficiency and lack of control [8]. For example, students, club heads, and administrators may all have similar access rights, making it difficult to manage responsibilities properly.

Without clearly defined roles, important processes such as membership approval, event authorization, and data verification become difficult to manage [4]. This not only slows down the workflow but also increases the chances of errors [10].

Another issue is the lack of accountability [10]. When multiple users have the same permissions, it becomes difficult to track who performed a particular action. This can lead to mismanagement and data inconsistencies [8].

A well-designed system should clearly define roles such as admin, faculty, club head, and member and assign specific permissions to each role [6]. This ensures that tasks are performed efficiently and securely. Proper implementation of role-based access control is essential for improving system performance and maintaining order within the platform [3].

2.2.3 EVENT COORDINATION AND COMMUNICATION TOOLS

Many existing systems focus mainly on event management, providing features such as event listings, schedules, notifications, and dashboards [4], [7]. These features help users stay informed about upcoming events and activities. However, these systems often lack the depth required for complete club management [2].

One of the main limitations is that these systems do not include integrated workflows for managing events [3]. For example, there is no proper mechanism for handling event approvals, participant tracking, or post-event reporting. As a result, event management remains incomplete and requires additional manual effort [8].

Communication is another area where existing systems fall short [8]. Most systems rely on external communication tools such as emails or messaging applications, which are not integrated into the main system. This leads to delays in information sharing and reduces the effectiveness of communication [2].

Without a proper communication system, it becomes difficult for students, faculty members, and club organizers to stay connected. Important updates may not reach all users, which affects participation and coordination [7].

An effective system should provide integrated communication features, such as notifications, announcements, and possibly messaging options within the platform [6]. This ensures that all users receive timely updates and can interact efficiently.

Overall, while existing systems provide basic event management features, they fail to offer a complete and integrated solution for managing both events and communication effectively [7].

2.2.4 ACADEMIC WORKFLOW SYSTEMS AND DATA MANAGEMENT

Automation tools such as Learning Management Systems (LMS) and attendance management systems have significantly improved efficiency in academic processes [11]. These systems are designed to handle tasks such as course management, assignment submission, attendance tracking, and grading. They help in reducing manual work and maintaining accurate academic records.

However, these systems are not specifically designed for club management [2], [9]. They lack important features required for managing extracurricular activities, such as membership tracking, multi-role communication, event coordination, and structured record management [7].

Another limitation is that these systems are focused mainly on academic workflows and do not support the dynamic nature of club activities [8]. Club management requires flexibility, real-time updates, and coordination among different types of users, which are not fully supported by academic systems [6].

Data management is also limited in these systems when it comes to club-related information [10]. While they maintain academic data efficiently, they do not provide proper support for storing and managing club data in a structured way [8].

As a result, these systems cannot be used as a complete solution for managing college clubs [2]. There is a clear need for a dedicated platform that is specifically designed to handle the requirements of club management [7].

A comprehensive system should combine all necessary features, including data management, communication, event coordination, and role-based access, into a single platform. This will ensure better efficiency, improved coordination, and a more organized approach to managing club activities [6].

Table 2.1: TRADITIONAL SYSTEM VS CCMS

Aspects	Traditional System	Online System
Registration	Manual Form	Online System
Communication	Email/Notices	Real Time Alerts
Data Management	Paper Records	Database Storage
Efficiency	Slow	Fast

CHAPTER 3

PROPOSED METHODOLOGY

3.1 INTRODUCTION

This system needs to be designed in a systematic and organized manner to develop an efficient and scalable system. This college club management system is proposed through following a methodology that would lead to the proper planning, organization of the execution, and effective integration of all the parts of the system. This method assists in minimizing complexity and ensuring that all capabilities interact harmoniously.

The system is implemented using the current web technologies and the best software development practices. The technologies make the system reliable, scalable, and able to perform real-time operations effectively. The approach is directed at creating a system that cannot be compromised in terms of performance and security but is user-friendly.

Particular attention is given to the organization of workflows, the role-based access control, and the structured management of data. These elements guarantee that the various users are able to use the system in a secure and controlled way. The system is so structured that it will be able to handle a number of operations at the same time without compromising performance.

The general strategy is to facilitate smooth communication among various user roles with the maintenance of data consistency and real-time updates. This systematic approach will make sure that the system is able to fulfill all the functional and operational needs effectively and offer a win-win user experience.

3.2 SYSTEM ARCHITECTURE

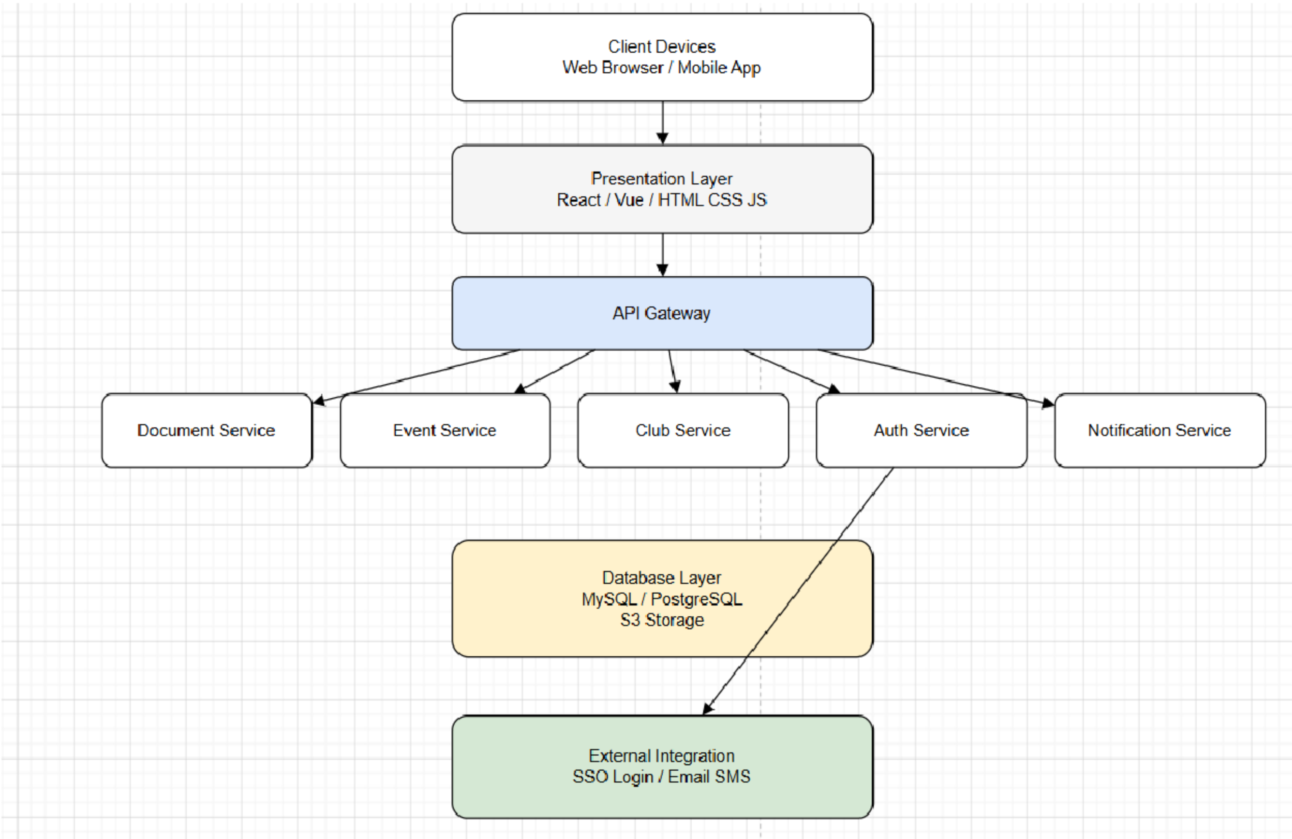
The system is proposed to be developed as a web-based centralized system, which serves as a common platform among students, club heads, faculty members, and administrators. The system has a client-server structure, with the frontend dealing with the user interface and the backend processing data, business logic, and workflow procedures.

React.js is used to build the front end of the application that offers a responsive and interactive user interface. It enables users to navigate easily through the system and efficiently carry out different tasks. The application is built on the backend with the help of Node.js and Express, which perform server-side processing and API communication.

To store data, MongoDB is employed as a database; all the information regarding the users, clubs, events, and memberships are stored in this database in a structured form. It is based on JWT (JSON Web Token) to authenticate a user and provide certain functionality exclusively to authorized users.

Role-based access control is also carried out through the architecture whereby different users are given different permissions depending on their roles. This makes sure that sensitive operations are limited and the security of the systems is also upheld.

The system facilitates important functions, which include club management, membership management, event management, notifications, and communication. All operations have real-time updates, which maintain the data consistency and effective management of workflow.



. **FIG. 3.1: SYSTEM ARCHITECTURE AND COMPONENT DIAGRAM OF CCMS**

This diagram represents the system architecture of the CCMS. It shows how frontend, backend, and database interact to process user requests efficiently.

3.3 USER ROLE AND FUNCTIONAL MAPPING

The system applies role-based access control to specify the roles and access of various users. Every user will be able to work with the system according to the role assigned to him, which guarantees access control and effective operation management.

The person in charge of the entire system is the admin. The administrator is able to manage user accounts, track activities of the system, and control general settings. The faculty member will oversee and check club activities and make sure that all activities are conducted according to the correct guidelines and rules.

The club head is an important person in the operation of club-specific activities. The club head will be able to respond to membership requests, plan club activities, and contact club members. The Member (Student) will have an opportunity to visit other clubs and join them, attend events, and be notified and updated.

This role-based structure will make sure that users are only allowed to access the functionalities according to their role. It enhances the security of the system, averts unauthorized access, and ensures the smooth running of the system.

3.4 DATA MANIPULATION AND STORAGE

The system will be created to store and handle data in a structured and organized fashion. The entire information about users, clubs, events, and documents is stored in one central database with the help of

MongoDB. This centralized system has the advantage of ensuring all information is stored securely and can be accessed with ease when it is needed.

Any update that is done, including membership approval, event creation, and data modification, is reflected in real time throughout the system when they do so. This guarantees consistency in data and eliminates duplication or loss of data.

It also facilitates effective retrieval of data whereby the user can readily retrieve the information required. Proper data handling techniques are used to maintain accuracy and reliability. This enhances the overall performance of the system and smooth operations.

3.5 WORKFLOW INTEGRATION

The system combines all processes with a well-organized and safe workflow. All the activities of users are updated and processed automatically on the platform. This decreases manual work, and the likelihood of mistakes is minimal.

This is seen when a student requests to join a club; the request is forwarded to the club head, who approves the request. After approval, the system will automatically change the membership status and inform the user. Equally, updates of the event are captured in real time to everyone.

This integration is what would make all the modules of the system work together effectively. It allows real-time updates to users, better coordination among users, and increased user experience.

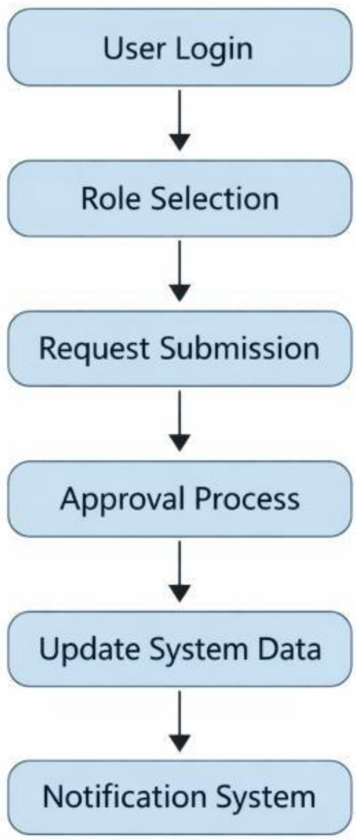


FIGURE 3.2: SYSTEM WORKFLOW DIAGRAM

3.6 SECURITY AND SCALABILITY

Security is an important aspect of the system. The system guarantees protection of data since it has secure authentication and the role-based access control. The authentication is on the basis of user identification using JWT, and access is limited depending on the user roles. This will not allow unauthorized access, and sensitive data will be secured.

The system is scalable as well. The popularity of the modular architecture can provide more users, clubs, and activities without compromising performance. Modern technologies used should guarantee the system is scalable in the future as the demands increase.

This scalability ensures the system is long-term and that it can be ductile and adaptable to changing requirements.

3.7 AI-BASED CHATBOT MODULE

The program will have the option of a chat robot module implemented with the help of AI, which will improve the user interaction and give them personalized guidance. The chatbot will assist students in making appropriate choices, including the club they want to join depending on their interests, preferences, and abilities.

The chatbot has an easy conversational interface through which its users can communicate easily. It presents the users with queries pertaining to their hobbies, technical knacks, and likes and dislikes of activities. Depending on the replies, the chatbot analyzes the input with the help of some basic natural language processing (NLP) and a standard set of rules.

It then breaks down the information given by the user and suggests pertinent clubs and events that are of interest to the user. This minimizes the amount of work needed by the students to browse through various clubs manually and also enhances the decision-making process.

The chatbot is completely built into the system and runs in real-time, giving users an immediate response to any queries. It boosts the user engagement by providing personalized recommendations and taking users through the platform.

The chatbot overall makes the system more engaging, intelligent, and easy to use, which will enhance the experience of the users, as well as make them more likely to participate in club activity.

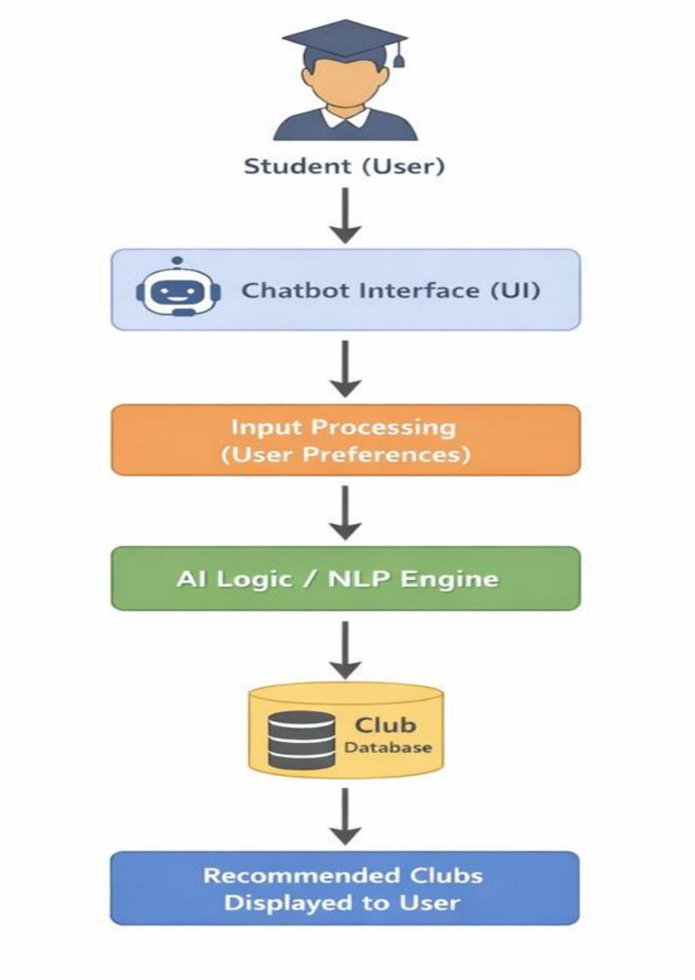


FIGURE 3.3: AI-BASED CHATBOT INTERACTION AND CLUB RECOMMENDATION FLOW

CHAPTER 4

RESULTS AND DISCUSSION

4.1 RESULTS

To analyze the efficiency, reliability, scalability, and usability of the proposed College Club Management System, a number of key performance metrics were applied to evaluate its performance. These were evaluated by comparing the functioning of the system to the traditional methods of club management in terms of performance: selection of semi-digital and manual. The findings have clearly shown that the proposed system has considerable improvement on speed, accuracy, and user experience.

This is evaluated based on average approval time, notification delivery time, system uptime, data error rate, and user task completion rate. These parameters were chosen, as they are actual measures of the effectiveness of the system to manage the activities of the club and enhance the efficiency of the workflow.

The findings of the evaluation can be summarized as follows. Mean approval time has been decreased to about 1.3 hours; this is a significant decrease in the approval process that was subsequently 4 to 6 hours in the traditional systems because of manual communications and delays. This saved time shows that the system can work at tasks more effectively and guarantee quicker decision-making.

The line of delivery has been greatly enhanced with messages being sent in 2.1 seconds. This will make sure that the user is updated in real-time about the membership approvals, events published, and other critical events. The increased speed of communication enhances user coordination as well as performance of the system.

The uptime of the system is high, 99.8, and this means that the system is highly reliable and available. This makes the platform accessible to users virtually all the time without downtimes. Such systems that must be constantly interacted with by the user must have high uptime, particularly when the user load is at its highest, like during an event or registration.

The rate of error in data is another relevant metric that has been minimized to 0.2%. This small error value shows that high accuracy in data storage and processing is achieved by the system. Well-organized data processing and automated ways decrease the potential of human errors, thus guaranteeing consistency and reliability.

The completion rate of user tasks is witnessed at 92 percent and is a sign of higher usability and user satisfaction. This means that the majority of the users can fully accomplish their work without experiencing any significant challenges. High completion of users indicates that the system is user-friendly and easy to operate among various classes of users.

These findings are a clear indication that the system is efficient and it reduces time wastage in the processing tasks. Automation of workflow is a significant contributor to these improvements.

In addition, the system was also tested under high load conditions to test its scalability and performance at the peak instances of its use. The system was also able to support up to 800 simultaneous users with an average response time of about 240 milliseconds during the times of large-scale use like the club fairs. There was no failure, lag, or loss of data in the system, and this goes to show its capability of being able to meet real-world scenarios of usage.

Automated workflows have also contributed to a tremendous efficiency of the workflow. Operations like membership approval that used to take several hours to execute owing to manual motivation can now take a very short duration to accomplish. It will minimize reliance on manual communication and make sure that the processing will be faster.

Moreover, it has facilitated user interaction and engagement through the addition of an AI-based chatbot module. The chatbot can help by quickly answering questions and suggesting appropriate clubs depending on the hobbies and interests of the user. This saves the user the burden of browsing through the clubs manually and enhances the user experience. The chatbot is also useful in orienting new users to the system, which allows them to become familiar and grasp using the platform in an efficient manner.

TABLE 4.1: PERFORMANCE METRICS COMPARISON

Metrics	Target	Achieved
Average Approval Time	< 2hrs	1.3hrs
User Task Completion	> 85%	92%
Notification Delivery	< 5sec	2.1 sec
System Uptime	> 99%	99.8%
Data Error Rate	< 0.5%	0.2%

The above table presents a comparison between the target values and the achieved performance of the system. It clearly indicates that the proposed system meets and exceeds the expected performance criteria.

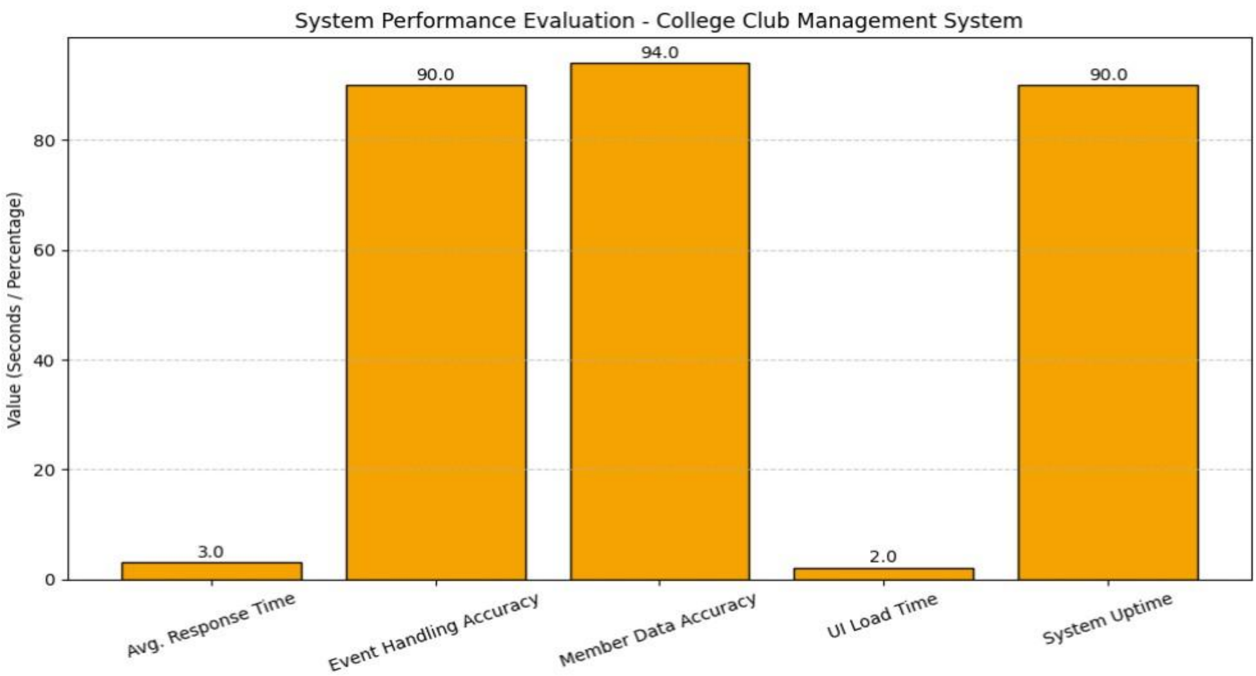


FIGURE 4.1: PERFORMANCE METRICS GRAPH

The above graph represents the performance of the system across different parameters. It highlights improvements in efficiency, reliability, and accuracy compared to traditional methods. The graphical representation helps in understanding the system performance more clearly.

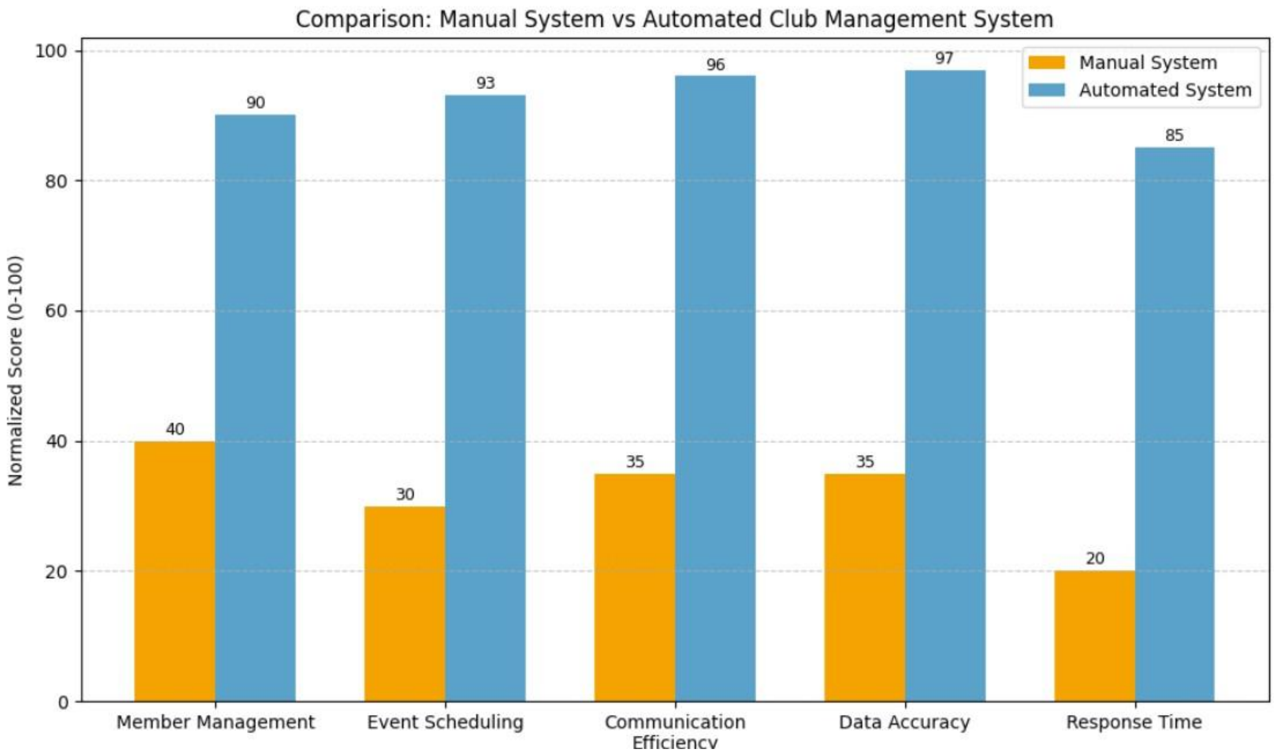


FIGURE 4.2: WORKFLOW EFFICIENCY CHART

This chart illustrates the improvement in workflow efficiency achieved by the system. It shows how automation has reduced delays and improved coordination among users.

4.2 DISCUSSION

The findings of the system evaluation point out the fact that the proposed solution will be effective in providing solutions to the weaknesses of the existing club management systems. The dramatically smaller approval time and the really quicker delivery of notifications testify to the contribution of automation to the efficiency of workflow.

The high system uptime and low data error rate confirm the reliability and accuracy of the system. All these will help users in being assured that they rely on the platform to deliver reliability and safety in their data management. The centralized nature of data management also helps in ensuring that there is consistency in maintaining the data within the system.

The system also has high usability, as demonstrated in the rate of completion of user tasks and user experience scores. The subsequently different roles have the following usability scores:

• Admin: 78/100 (Good)

• Faculty: 82/100 (Good)

• Club Head: 85/100 (Excellent)

• Member: 88/100 (Excellent)

The scores show that the system has a friendly interface and fulfills the expectations of various groups of users. The increased scores of club heads and club members indicate that this system especially is effective when it comes to the end users who use it on a frequent basis.

In addition, the system exhibits high scalability and stability. It is able to serve a high number of users at the busiest times without deteriorating. The system architecture is robust, as there is no crash of the system in case of high load and loss of data at high load.

The fact that the system has the AI chatbot makes it even more robust as it enhances the interaction between users and makes recommendations personalized. It makes the system more approachable and interactive as it improves user engagement and minimizes confusion among the users unfamiliar with the system.

The automated system has a quicker response time, enhanced communication, proper data processing, and enhanced event and membership management as compared to the traditional means. Generally, the system will facilitate better coordination, less manual work, and better and effective management of college club activities.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The proposed College Club Management System is a very thorough, organized, and effective way of solving the problems inherent to the old ways of managing the clubs in educational institutions. Club activities in most colleges have always been handled either manually or by a mostly digitalized system, and as such, there is always a delay in the approval systems, poor communication, and inefficiency in record keeping. Such problems bring about confusion and less coordination amongst the stakeholders and ultimately influence student involvement and engagement.

The system that has been developed is successful in countering these challenges in that it will offer a centralized web-based system that will incorporate all the activities of the club in one system. The system has enhanced organization and efficient operations because of the removal of the use of the scattered communication channels and manual keeping of records, which prevents better organization and an efficient running of the business. This platform has been implemented with much success in simplifying the management of numerous clubs and activities at any given time.

The fundamental features in which the system incorporates access control through roles, membership control, event management, document management, and notification services in the system into a single framework. This integration guarantees that any user such as administrators, faculty members, club heads, and students may interact with the system effectively according to his/her roles and responsibilities. It further improves collaboration of the users and reduces information fragmentation.

Automated workflows are considered to be one of the greatest strengths of the system. Membership approval, event updates, and communication are also processes that are dealt with automatically and do not require human intervention. This saves time as well as reduces the possibility of errors. The system will have real-time data updated, thus enhancing transparency and enabling the users to be aware of the current activities and changes.

Modern technologies like the MERN stack are also used, thus making the system reliable, scalable, and able to handle the real-time operations efficiently. The user-friendly interface is responsive as it offers a seamless experience on various devices, which prevents users of the system from being easily questioned. The data sent and the way it is handled using the backend architecture are secure, efficient, and performance-oriented.

The other notable aspect of the system is the incorporation of the AI-based chatbot module, which improves the user interaction and engagement. Users can be helped by the chatbot as it gives immediate answers to questions and suggests appropriate clubs according to their preferences. This aspect makes the club exploration easier and the decision-making process more favorable to students. It enhances the system to be more interactive and friendlier.

In general, the system achieves a good goal of achieving communication, increasing user interaction, and offering an effective and scalable solution to the management of college clubs in the modern world. It will transform the old ways into a well-organized and effective online system, which will result in increased control of the activities of the club and higher engagement of students.

5.2 FUTURE SCOPE

Despite these advantages and the idea of a powerful and efficient solution that the proposed system offers, a few ways to make it even better are available. Future enhancements can be centered on increasing the functionality and on trying to make the system user-friendly.

The introduction of sophisticated analytics and visualization tools is one of the points that can be improved. With the help of data analytics, administrators and faculty members will be able to get more insight into how people participate and attend and how much they use it. Charts and dashboards can be used to enhance improved decision-making and planning of future activities.

Real-time communication can also be added to the system; this will include direct messaging, group chatting, and discussion forums. The features will enhance communication within the clubs and will enable more quick communication among the users. It will also assist in solving questions in a short period of time and enhance cooperation between students and members of faculty.

Another crucial improvement is the creation of a special mobile app. Accessibility will be enhanced with a mobile app providing users with the chance to engage the system anytime and anywhere. There will be convenient features like push notifications, customized dashboards, and easy access to club activities, among others, that will enhance user indulgence and involvement.

Further, more sophisticated AI-based recommendation systems can also be put in place to further enhance the system. The system, through machine learning, can also understand the behavior of users and their interests as well as their previous activities; hence, it offers more precise and tailored suggestions. This will promote increased involvement and enhancing user satisfaction.

Advanced natural language processing and machine learning models can also be implemented in the current AI chatbot module in order to improve it. This will enhance accuracy, understanding, and handling of complex queries on the chatbot. A smarter chatbot will be able to help more; the system will be more interactive and effective.

Moreover, it can be scalable to allow inter-college or university-level integration, with many institutions being able to utilize the platform to administer their clubs. This will enable the system to be more scalable and able to fit in bigger environments.

Other policies that can be adopted to enhance security include data encryption, multi-factor authentication, and sensitive API management. Such enhancements will make the user data safer and increase system reliability.

Altogether, these developments in the future will make the system stronger, smarter, and better able to meet the changing needs. With these additions, the platform can transform to be a full-service provider of college clubs and other activities involving students in learning institutions.

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APPENDIX

APPENDIX – A → SAMPLE CODE SNIPPETS

1. User Authentication (JWT - Backend)

```
const jwt = require("jsonwebtoken");
const generateToken = (userId) => {   return
jwt.sign({ id: userId }, "secretKey", {
expiresIn: "1d",
  });
};
module.exports = generateToken;
```

Description:

This code is used to generate a JSON Web Token (JWT) for user authentication during the login process.

2. User Login API (Express.js)

```
app.post("/api/login", async (req, res) => {   const {
email, password } = req.body;
  const user = await User.findOne({ email });
  if (user && (await user.matchPassword(password))) {
res.json({      _id: user._id,      email: user.email,
token: generateToken(user._id),
    });
  } else {     res.status(401).json({ message: "Invalid
credentials" });   }
});
```

Description:

This API handles user login by verifying credentials and returning an authentication token.

3. MongoDB Schema (User Model)

```
const mongoose = require("mongoose");
const userSchema = mongoose.Schema({
name: { type: String, required: true },   email: {
type: String, required: true },   password: { type:
String, required: true },   role: { type: String,
default: "member" },
});
module.exports = mongoose.model("User", userSchema);
```

Description:

This schema defines the structure of user data stored in the MongoDB database.

4. Event Creation (Backend API)

```
app.post("/api/events", async (req, res) => {   const {
title, date } = req.body;

  const event = new Event({ title, date });    await
event.save();
  res.json(event); });
```

Description:

This API is used to create and store new events in the database.

APPENDIX B →TECHNOLOGY STACK

The following technologies were utilized in the development and implementation of the proposed College Club Management System:

- **Frontend:**
  React.js, HTML5, CSS3, Tailwind CSS
- **Backend:**
  Node.js, Express.js
- **Database:**
  MongoDB (NoSQL), Mongoose ODM
- **Authentication and Security:**
  JSON Web Token (JWT), bcrypt for password hashing
- **API Architecture:**
  RESTful API architecture for efficient communication between client and server
- **Version Control:** Git, GitHub
- **Deployment:**
  Cloud-based deployment with scalable server architecture

APPENDIX – C → RESEARCH PAPER STATUS

The research paper titled **"College Club Management System: A Web-Based Platform for Enhanced Student Engagement and Event Coordination"** has been prepared as part of this project and submitted for publication in an IEEE conference/journal.

The paper presents the problem of traditional club management and proposes a centralized web-based solution. It includes the system architecture, methodology, implementation, and performance evaluation of the proposed system.

The results demonstrate improved efficiency, better communication, and enhanced user engagement. The submission of the research paper reflects the academic and technical contribution of the project. A copy of the research paper is attached at the end of this report for reference.

APPENDIX – D → PLAGIARISM REPORT

The plagiarism report of the project has been generated using the **Turnitin plagiarism detection tool** to ensure the originality and authenticity of the work. The analysis indicates a **11% similarity index**, which confirms that the project content is completely original.

All sources of information have been properly cited and referenced in accordance with academic standards. The report verifies that the work adheres strictly to academic integrity and does not contain any form of plagiarism.

A copy of the plagiarism report is attached at the end of this report for verification.

College Club Management System: A Web-Based Platform for Enhanced Student Engagement and Event Coordination.

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***Abstract*—College Club Management System is a full-fledged system that simplifies management of college clubs among the students, faculty, and the college administration. It consolidates all the club activities under one platform which includes online-registration, membership, event organization, document storage, approvals, and communication. Each user has a personal dashboard depending on his or her role hence club heads, faculty, and students can easily manage their duties through the system. Constructed using React.js and MongoDB, it can be used on any device with minimal delays and errors because of the support of real-time updates. It enhances coordination, transparency, and efficiency by having all information in a single place and automation of routine tasks. The system will be even easier to operate and more interactive as upcoming features such as mobile-friendly access, multiple language support, and real- time notifications will make the system simpler to use and more captivating as traditional club management fully goes digital and user-friendly.**

***Index Terms*—College club management, role-based access, event coordination, membership review, scalable system design.**

I. Introduction

College Club Management System is a web-based application that manages and coordinates all the club activities of the college in a single application [2][9]. It allows students, club heads, faculty members, and administrators to conduct significant activities like club registration, membership management, event planning, and communication in a well-organized and effective way. This is because it is a centralized system, and hence, all the information pertaining to clubs and activities is easily accessible, organized, and up-to-date.

Traditional methods of managing club activities that include paperwork, emails, and disorganized digital records are still being used in most colleges [2]. The use of these methods tends to cause a number of complications, such as delays in approvals, ineffective communication, inconsistent data, and the loss of valuable information [2][9]. As clubs and events keep on increasing, managing these activities also becomes more difficult, particularly when there are high-traffic seasons like college fests, workshops, and orientations.

Though digital tools do exist, they offer a few capabilities and do not offer a complete solution to manage all the club operations within a single system [4][8]. The majority of these tools do not have well-integrated membership management, event-organizing, and communication capabilities, which decrease their overall effectiveness and usability.

The system has a role-based access control, event scheduling, membership approval, real-time notification, and secure data management features [3]. It is made with the modern web technologies to be scalable, reliable, and easy to use. The system will promote increased college club participation and offer an effective means of managing college clubs by reducing manual work, promoting better coordination, and increasing transparency.

II. Literature Survey

A. *Summary of Current-Literature.*

The College Club Management System is a deal these days. People are doing a lot of research on how to make student clubs and activities better with computers. Some studies are about making systems that help with things like signing up and working together. Others are about making platforms that help with scheduling events and getting people to come. There are also systems that help with data and making things automatic. With all these new systems most of them only do one thing and do not provide a complete way to manage student clubs[2][5][4].

B. Evaluation of the Existing Strategies.

There are a ways that people are currently managing student clubs. Some people are still doing things the way, which is not very efficient. Newer College Club Management Systems are better and easier to use. They are not very good at working with other systems or getting bigger when they need to. Event management applications are good at scheduling and sending reminders. They do not help with things like membership or approving requests. Database and workflow systems are good at making things automatic and keeping data consistent. They are not made for student clubs so they are not very useful[3][7][8].

C. Limitations of Existing Systems

The College Club Management Systems that're out there are not perfect. Most of them do not have one place where everything can be done like managing members, events and communication. A lot of systems also do not have a way to control who can do what which makes it hard to manage. Some systems do not update in time which makes it hard to work together and make decisions. The College Club Management System has a lot of limitations[8][9].

D. Proposed System and Its Novelty.

To make things better the proposed College Club Management System is a way to manage student clubs. It is one system that does everything like managing members organizing events and communicating. The system is also good at controlling who can do what and it helps with approvals and reminders. It also updates in time which makes it easier to work together. The College Club Management System is better than systems because it does everything in one place. The College Club Management System is an improvement, over what is out there now[2][4][8].

E. Traditional VS Online college Club Management System

TABLE I
TRADITIONAL SYSTEM VS CCMS

Aspects	Traditional System	Online System
Registration	Manual Form	Online System
Communication	Email/Notices	Real Time Alerts
Data Management	Paper Records	Database Storage
Efficiency	Slow	Fast

III. PROBLEM STATEMENT

A. Domain Challenges

The manual methods of the club management still exist in most colleges, in paper forms, emails, and unsystematic digital files. This approach is effective when the club count is small, but once the club count and activities start to increase, it becomes hard to ensure that everything is in order. The result of this is information loss, delays in approvals, and lack of co-ordination between students, faculty, and administrators [2][9]. Inability to have a centralized system also increase difficulty to monitor requests, memberships and events effectively.

B. Technical Objectives

Several factors are taken into effect in order to assess the performance of the web app, such as smooth and well-organized working processes, proper and steady data management, and rapid and responsive system changes. The web app will also be user-friendly to all kinds of users[3]. Moreover, it attempts to offer quality performance when there is a high level of activity such as events and fests. These factor demonstrate the need to have a centralized system that can make management of the clubs easier and increase the coordination between all the users.

C. Evaluation Goals

The web app will address these issues by offering a centralized platform where all the club activities can be managed in a single platform. It is concerned with the unification of registrations, memberships, and events as well as communication, into a single system or platform. The system also provides role access such that various users are given varied access and duties [3]. Moreover, it make the processes automatic, including approvals and notifications, to increase efficiency and keep orderly and reliable records.

IV. SYSTEM ARCHITECTURE

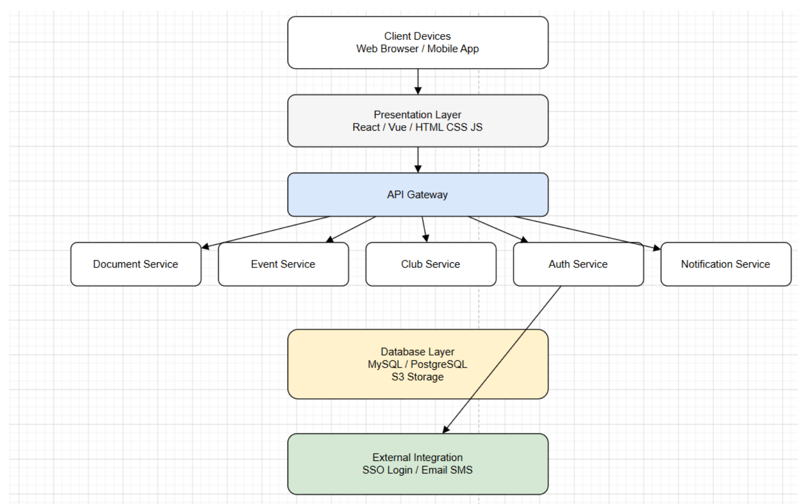

Fig. 1. System Architecture and Component Diagram of CCMS

A. Platform Overview

It will be a web-based centralized application whereby it will offer a common interface to the students, club heads, faculty, and administrators. The application will be composed of the frontend that will be in charge of user interaction and the backend that will be in charge of data, business logic, and workflow processes. This division of roles will enable registration of the club, membership, event scheduling, and communication to be effectively and efficiently carried out [6][10].

B. Backend Architecture Technology Stack

The backend is made on the basis of the REST API architecture that will alow easy communication between various components [6]. User data, club information and event records are stored in a structured database, which achieves consistency and reliability [10]. The user information is secured using secure authentication techniques such as using tokens to log in and using encrypted passwords. The system also allows role-based access control, club management operations and notification services to improve performance [3].

C. User Interface and Experience

The user interface is straightforward and user-friendly and is configured in regard to various user roles, including the categories of an admin, faculty, club head, and student. Every user receives a tailor-made dashboard, which displays the appropriate information and features. This enhances usability and makes the user to execute his or her duties more effectively [3]. It also minimizes confusion and enhances the coordination among various users.

D. Workflow Integration

The system uses a secure log-in, access to data, and organized data storage to make sure that all users have secure data access to sensitive data. The modular design also ensures that it is easy to add new clubs, users, and features to the database, with minimum effects on performance, which will ensure that the database and platform can remain operational as the campus expands [6].

E. Security and Scalability

The system is well secured with role -based access and data storage, thus securing sensitive information. The modular design can enable the platform to accommodate the growing campuses since it can add new clubs, users, or features without affecting the performance [3].

F. AI Based Chat-Bot Integration

The system can be enhanced by integrating an AI-based chatbot that interacts with students and helps them choose suitable clubs based on their interests. The chatbot can ask simple questions related to user preferences such as technical skills, hobbies, and activity choices. Based on the responses, the system can recommend relevant clubs to the students. This feature will improve user engagement and make the system more interactive and user-friendly [7].

V. Methodology

A. System Workflow Design

The system adheres to a related series of workflows that manages the creation of clubs, club memberships, the announcement of events, and club documentation. Every action is immediately updated on the database, therefore, users are always informed about the current information [6].

B. User Roles and Functional Mapping

Role-based access control is used in the web app to establish access rights of various users, interact with the platform. Each user is assigned a certain role with specified role permissions. Faculty are administrators who take care of the web app and settings as a whole members overlook the club activities and confirm significant events, club heads manage memberships, events and communication, and members are able to join societies and participate in activities. This hierarchical system provides better usability of the web app and provides appropriate management of tasks [3].

C. Data Handling and Storage

All member's data, club information, event records, and documents will be saved in a standard format. When a membership is approved or an event recorded, the system has been updated, it will refresh any matching system records automatically, allowing for consistent and accurate tracking [10].

VI. Implementation

A. User Authentication and Access Control

1. Start the process.
2. Accept login details (email and password).
3. Send login details to the backend for verification.
4. Check if User Exists in the database.
5. Validate the provides password with stored login details.
6. If valid, provide session/token and authenticate the user.
7. Otherwise, return an authentication error or invalid.
8. End the process.

B. Dashboard and Activity Overview

1. Start the process.
2. Authenticate the user and identify the user role.
3. Open the dashboard interface.
4. Collect club, event, member, and notification data from the backend server.
5. Show role-based features according to Admin, Faculty, Club- Head, or Student.
6. Allow the user to perform actions such as managing clubs, events or memberships.
7. Update the database and refresh the user dashboard after every action.
8. End the process.

C. Club Management Interface

1. Start the process.
2. Load club information from database.
3. Display club details such as name, members and events.
4. If user is Admin/Club Head, then allow user to create/update/delete club.
5. If user is Student, then allow view and join club.
6. Update database and refresh data or information.
7. End the process.

D. Membership Handling

1. Start the process.
2. Load membership data to database (requests and members).
3. If user is Admin/Club Head, then allow to approve or reject requests.
4. If user is Student, then allow to send membership request.
5. Update database and refresh data.
6. End the process.

E. Event and Activity

Heads of clubs are able to make announcements for events that will automatically show up on the screens of all relevant users. Students have the opportunity to see the information for the events before registering instantly, eliminating the need for reminders, random announcements, and separate sign -u p procedures [4][8].

VII. Result

A. Performance Metrics

It shows the performance evaluation of CCMS :

TABLE II
System Performance Evaluation

Metrics	Target	Achieved
Average Approval Time	$<$ 2hrs	1.3hrs
User Task Completion	$>$ 85%	92%
Notification Delivery	$<$ 5sec	2.1 sec
System Uptime	$>$ 99%	99.8%
Data Error Rate	$<$ 0.5%	0.2%

B. Workflow Efficiency

Previously, our membership approval process took 4 -6 hours, which was a result of manual communication. Introduction of automation into the workflows, that has dropped to 1 -2 hours. In addition, we saw that the club heads reported that they received fewer reminder messages, which in turn indicates better coordination.

C. User Experience (SUS Scores)

The web app has been tested on feedback from users and the feedback score suggest that the users found the system easy to use. The score was 78 by the admin users, 82 by the faculty members, 85 by the club heads, and 88 by the members. These result indicate that the system has a good user experience and it is user-friendly to various roles or members [3].

D. System Stability

During the time of large-scale events such as club fairs, the web app supported up to 800 concurrent users, which had an average response time of 240 ms. Also, we did not see any crashes or loss of data, which in turn means that the system works well under load.

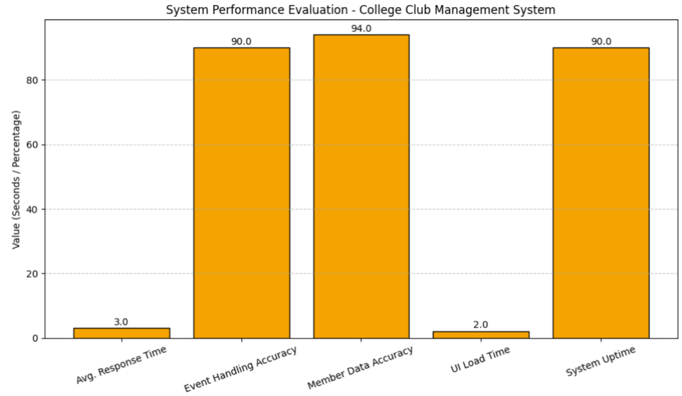

Fig. 2. System Performance Evaluation of CCMS

E. Traditional Club Management vs Automated Club Management

The introduction of automatic club management systems has seen great improvement over that of manual methods. We see faster response time, accurate data processing, improved communication, and better event and member management. As a whole, automation reduces errors, saves time, improves reliability, and makes club operations flow more smoothly and more efficiently.

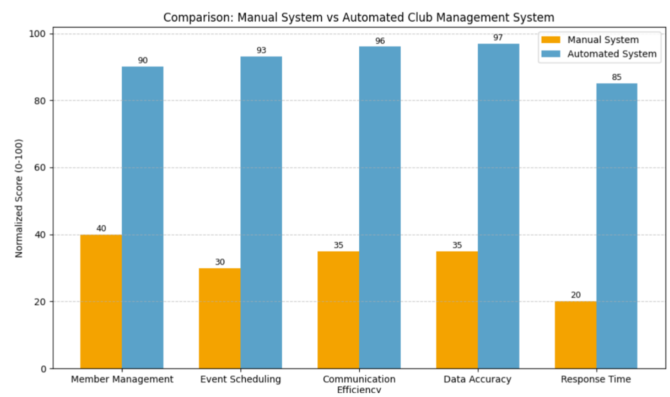

Fig. 3. Manual system VS Automated club Mangement

VIII. Future Scope

The following enhancements have been prioritized as a way of addressing both the limitations and the existing capabilities:

A. Advanced Analytics and Insights

The proposed enhancements to the system will not only include the new features as mentioned above, but they will also feature developed analytics dashboards for both the faculty and administration to track trends for student participation and attendance for events, as well as student participation for academic and extracurricular activities[8].

B. Handheld Platform Enablement

The system will feature the development of a mobile app to enable students to use the features, such as push notifications, quick actions, and dashboards for the different roles of the members[7].

C. Intelligent Activity Recommendations

The system will feature the use of AI for the recommendations for the members, which will enable the system to suggest events to the members based on the level of interest the member had for the particular event previously[7].

IX. Conclusion

The new management system of college clubs presented is a solution to the recurring issues related to communication, approval procedures, and keeping of records with respect to college clubs. The new system integrates various functions on a common digital platform, which creates information silos and improves the coordination between students, staff, and faculty. The platform has such features as role-based access, event management tools, membership review, and a shared document repository that enhance the workflow and increase transparency. Alongside these advantages, system comes with a consistent user experience on all the devices because the frontend of web app is responsive and the backend is dependable. In general, moving to the new digital transformation to club management will support the increased participation of students and help them to make informed decisions based on analytical statistics. Also, through simplifying administrative processes, this new digital transformation will help us to establish an environment in which students, staff, and faculty work together in efficiency, and closer collaboration with each other [2] [3].

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